W. Blake Drechsler

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EDUCATION

UNIVERSITY OF VIRGINIA | *PhD, Astronomy* Expected May 2027
- Research Advisor: Dr. John Tobin

UNIVERSITY OF VIRGINIA | *MS, Astronomy* May 2023
- Research Advisors: Proff. Jonathan Tan and Dr. John Tobin

UNIVERSITY OF CALIFORNIA BERKELEY | *BA in Astrophysics* May 2020
- Research Advisor: Proff. Jessica Lu

RESEARCH EXPERIENCE

UNIVERSITY OF VIRGINIA

Characterizing Protostellar Masses and Mass Assembly Summer 2022-Present
- Advisor: Dr. John Tobin
- Analyzed JWST MIRI MRS data of the edge-on, Class 0 protostar L1527 IRS in order determine its accretion mechanism and accretion rate
- Performed radiative transfer modeling of the ALMA observations of Class 0/I protostars from the eDisk program

Simulations of Giant Molecular Cloud Collisions Summer 2021-Summer 2022
- Advisor: Proff. Jonathan Tan
- Used the `ENZO` MHD to simulate the collisions of GMCs to investigate if these events could be triggers of massive star and star cluster formation
- Worked on developing subgrid models to include the effects of stellar feedback

UC BERKELEY

Stellar Orbits in the Galactic Center Summer 2020-Summer 2021
- Advisor: Proff. Jessica Lu
- Used Bayesian inference techniques, such as nested sampling, to assist in characterizing the orbits of stars around Sgr A*

Keck Adaptive Optics (AO) Simulations Spring 2019-Summer 2020
- Advisor: Proff. Jessica Lu
- Ran and analyzed performance simulations of the next generation Keck AO system (KAPA)
- Helped determine the number and and arrangement of laser guide stars for the KAPA asterism

TEACHING EXPERIENCE

UNIVERSITY OF VIRGINIA

Astronomy 5470 TA Spring 2026
- Graduate-level computational astronomy course
- Assisted the students with in-class exercises and helped grade weekly assignments.

Astronomy 3130 TA Spring 2023
- Observational astronomy course for third and fourth year majors
- Assisted in running the labs, taught a number of lectures, and assisted students through office hours and workshops.

Telescope Observing TA Fall 2021
- Lab at McCormick Observatory required for many entry level astronomy courses
- Assisted students during the lab, helped undergraduate TAs operate the telescopes, set up and closed down the facility, and graded the labs

Astronomy 1210/1220 TA Fall 2021
- Introductory Astronomy courses on the Solar System (1210) and Stars, Galaxies, and the Universe (1220)
- Assisted students via office hours and one-on-one meetings, led review lectures before exams, and graded assignments/exams.

PRESENTATIONS

Invited Talks

Virgnia Tech Astronomy Seminar September 2025
- Blacksburg, Virginia
- *"Studying How Stars Grow in the Earliest Stages of Protostellar Evolution"*

Lorenz Workshop: Protostellar Accretion and Ejection with JWST and Beyond May 2025
- Leiden, Netherlands
- *"Accretion in L1527 IRS"*

Contributed Talks

Missing Links in Planet Formation November 2026 (Upcoming)
- Taipei, Taiwan
- *"Obtaining a Sample of Protostellar Masses via Radiative Transfer Modeling"*

Posters

Discs on the Exe July 2026
- Exeter, UK
- *"Robust Stellar Mass Estimates for the Edge-On Class 0 Protostar L1527 IRS"*

Protostars and Planets VII April 2023
- Kyoto, Japan
- *"Determining the Mass and Accretion Mechanism for the Class 0 Protostar L1527 IRS"*

From Stars to Galaxies II Summer 2022
- Gothenburg, Sweden
- *"Massive Star Clusters from Colliding Giant Molecular Clouds"*

Our Galactic Ecosystem Spring 2022
- Lake Arrowhead, California
- *"Massive Star Clusters from Colliding Giant Molecular Clouds"*

Keck Science Meeting Summer 2020
- Online meeting
- *"Performance Simulations of the Keck All-Sky Precision Adaptive Optics System (KAPA)"*

ACCEPTED OBSERVING PROPOSALS

PI: ALMA Cycle 13
- When the Dust Settles: Measuring the Dust and Gas Scale Heights for a Sample of Class I Protostellar Disks (A-rank 13.5 hours)

SKILLS

Programming Languages Python, C/C++, Latex

Software Git, CASA, RADMC3D, Unix

Python Packages numpy, matplotlib, astropy, emcee, dynesty, scipy, photutils

Analysis Methods Radiative transfer, Bayesian inference (MCMC, nested sampling)

AWARDS / FELLOWSHIPS

GROTE REBER DOCTORAL FELLOWSHIP 2026-2027

- National Radio Astronomy Obsevatory (NRAO)
- Fellowship given to PhD students to conduct research in radio astronomy, radio instrumentation, or computational techniques at one of the NRAO sites in collaboration with an NRAO staff astronomer or engineer

DEAN'S LIST Spring 2019

- UC Berkeley
- Award given to students with a semester GPA in the top 10% of Letter and Sciences undergraduates who take at least 13 graded units and have no disqualifying grades that semester.

PUBLICATIONS

1. **Drechsler W.Blake**, (+ 14 co-authors), 2026, "Accretion onto the Embedded Protostar L1527 IRS: Insights from JWST NIRSpec and MIRI Observations", ApJ, 1000, 299

2. Devaraj, R. (+13 co-authors and **Drechsler, W.B**), 2026, "JOYS: JWST MIRI/MRS spectra of the inner 500 au region of the L1527 IRS bipolar outflow", arXiv:2601.17820

3. Slavicinska, K. (+16 co-authors and **Drechsler, W.Blake**, 2025, "HDO Ice Detected toward an Isolated Low-mass Protostar with JWST", ApJL, 986, 2

4. Finn, M., Mazzei, R., **Drechsler, B.**, Telkamp, Z., Rao, M., Agrawal, P., McAlister, 2023, "How Informal Science Education Influences Elementary Students' Perceptions Of Science And Themselves." Journal of STEM Outreach. 6.10.15695/jstem/v6i1.10.

5. Jia, S., (14 co-authors + **Drechsler, W.B**, 2023, "Stellar Populations in the Central 0.5 pc of Our Galaxy. III. The Dynamical Substructures", ApJ, 949, 1